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Smart-Attend: An AI/ML Based Automated Attendance System

Koushiki Ghosh¹, Debolina Saha², Shubhrakamal Saha³, Prarambhika Bhattacharjee⁴, Sudip Dogra⁵

- ¹ B. Tech Student, Department of Electronics and Communication Engineering, Meghnad Saha Institute of Technology, Kolkata, India, Email: koushikig91@gmail.com
- ² B. Tech Student, Department of Electronics and Communication Engineering, Meghnad Saha Institute of Technology, Kolkata, India, Email: saha.debolina25@gmail.com
- ³ B. Tech Student, Department of Electronics and Communication Engineering, Meghnad Saha Institute of Technology, Kolkata, India, Email: shubhrokamalsaha04@gmail.com
- ⁴ B. Tech Student, Department of Electronics and Communication Engineering, Meghnad Saha Institute of Technology, Kolkata, India, Email: praram2005@gmail.com
- ⁵ Professor, Department of Electronics and Communication Engineering, Meghnad Saha Institute of Technology, Kolkata, India, is the corresponding author. Email: dogra.sudip@gmail.com

Abstract: Traditional attendance systems in educational institutions are often manual, time-consuming, and vulnerable to manipulation such as proxy marking. To overcome these limitations, this paper presents a smart attendance system powered by artificial intelligence (AI) and machine learning (ML), leveraging facial recognition to enable accurate and real-time attendance tracking. The proposed system captures student faces from video frames and instantly updates attendance records, ensuring seamless operation without manual input. A major strength of the system lies in its ability to detect and recognize masked or partially visible faces—commonly encountered in classrooms where one student may block another. By focusing on distinct visible facial features like the eyes and forehead, the system maintains high accuracy even in such constrained environments. Designed for flexibility, the solution performs reliably across varying distances, making it suitable for both small and large classrooms. Long-range face recognition capabilities allow it to function efficiently in auditoriums or wide lecture halls without compromising accuracy. To validate the system's performance, the model's recognition accuracy has been thoroughly tested and benchmarked. A user-friendly interface and robust database integration demonstrate its reliability and usability in real-time applications. This system not only automates attendance but also enhances transparency, security, and efficiency in academic administration.

Keywords—YOLOV10, MTCNN, FaceNet, SVM, KNN, Face Recognition, AI/ML

I. INTRODUCTION

A smart attendance system leverages advanced AI-based facial recognition technology to automate and enhance attendance tracking.

- 1) **Face Detection (YOLOv10):** Face detection locates and isolates faces for accurate attendance marking. YOLOv10 handles this as a single regression task, offering fast, accurate detection ideal for real-time recognition.
- 2) **Face Alignment (MTCNN):** Face alignment ensures that facial features (eyes, nose, mouth) are properly positioned before recognition. MTCNN (Multi-Task Cascaded Convolutional Networks) is used for the same.

It comprises three stages:

- (i) **P-Net (Proposal Network):** It provides windows that scan through a shallow CNN.

Steps:

- **Feature Extraction:** $F = w * I + b$

Where w = convolutional kernel (weight matrix), b = bias term, and I is the input image vector.

Convolutional layers are used to extract feature maps, which are then passed through the ReLU activation:

$$F' = \max(0, F)$$

- **Face Classification: (Binary Classification)**

It is done to determine if a region contains a face using a sigmoid function.

$$p = \sigma(W_f F' + b_f) = \frac{1}{1 + e^{-(W_f F' + b_f)}}$$

Where p = probability of the region containing a face.

- **Bounding Box Regression:** To improve face localization, bounding box offsets are predicted using:

$$\Delta x = W_r F' + b_r$$

Where Δx = adjustments to the detected bounding box.

- **Non-Maximum Suppression (NMS):** To remove duplicate detections.

- (ii) **R-Net (Refinement Network):** Refines candidate face regions from P-Net. It also uses convolutional layers, followed by fully connected layers.

- **Face Classification (Refinement)**

- **Cross-Entropy Loss:**

$$L_{cls} = -y \log(p) - (1 - y) \log(1 - p)$$

Where y is the ground-truth label (1 for face, 0 for non-face).

- **Bounding Box Regression (Improved Localization):**

$$\Delta x' = W_r' F + b_r'$$

- (iii) **O-Net (Output Network):** Refines detection and estimates five facial landmarks.

- **Landmark Localization:** Estimates five landmark points: left eye, right eye, nose, and mouth. Predicted using mean square error (MSE) loss.

$$L_{landmark} = \sum_{i=1}^5 (x_i - x'_i)^2$$

Where:

x_i = predicted landmark coordinates.

x'_i = ground-truth coordinates.

▫ *Final Classification and Regression:*

$$L_{total} = \lambda_1 L_{bbox} + \lambda_2 L_{landmark}$$

Where λ_1 and λ_2 are weights controlling the influence of bounding box and landmark losses.

- 3) **Face Recognition (FaceNet):** Face recognition verifies or identifies individuals by converting faces into 128D embeddings using FaceNet. Unlike pixel-based methods, it uses a CNN with triplet loss to ensure embeddings of the same person are close, and those of different people are far apart.
- 4) **Classification (SVM/KNN):** SVM and KNN are two commonly used classification algorithms. SVM works by finding the **optimal hyperplane** that separates data points belonging to different classes in a high-dimensional space. In KNN classification, a new data point is assigned a class based on the **majority vote of its K nearest neighbors**, where **K** is a user-defined parameter.

II. RELATED WORKS

[1] The paper introduces FaceNet, a unified system for face recognition, verification, and clustering. It maps face images to a 128D Euclidean embedding space where similar faces are close and different ones are far apart. Using triplet loss with anchor, positive, and negative samples, it minimizes intra-class distance and maximizes inter-class distance by a margin α .

[2] The paper presents an intelligent Raspberry Pi-based attendance system using deep learning for face detection, tracking, and recognition. It supports real-time and offline modes, includes a web interface, and uses centroid tracking and sharpness filtering for accurate verification in low-resource settings.

[3] The study presents a facial recognition-based attendance system using HOG for detection and SVM for recognition on a Raspberry Pi 3B+. It addresses lighting, distance, and orientation, performing best at 300–400 lux with frontal face alignment, and shows minimal impact from distance in classrooms.

[4] The Haar Cascade-based system uses OpenCV2 on NVIDIA Jetson Nano for efficient face detection and tracking. It automates attendance with 128D face vectors, real-time notifications, encryption, and a secure database, ensuring accuracy and privacy for schools and workplaces.

[5] The contactless attendance system uses dlib-based face recognition with a CNN-LSTM anti-spoofing module, achieving 95.85% accuracy. Integrated with a mobile app and Firebase, it ensures secure, real-time tracking and reduces storage by 33.52%, addressing biometric risks effectively.

[6] The paper proposes a multi-task cascaded CNN (P-Net, R-Net, O-Net) for joint face detection and alignment, improving accuracy by refining detections and predicting facial landmarks. It handles occlusions, pose, and lighting using adaptive hard sample selection and multi-loss training.

[7]. The paper “Dual-Proxy Modeling for Masked Face Recognition” proposes a method using two fully connected proxy layers to model masked and unmasked face features separately. It includes a balanced input module, CNN-based feature extractor, and dual-proxy module to ensure intra-class compactness and inter-class separation. An adapted ArcFace loss with cross-space constraints and gradient cutoff improves robustness. MaskTheFace augmentation aids training, enhancing recognition accuracy and generalization for both masked and unmasked faces.

[8] The paper presents a real-time attendance system using deep learning-based face recognition. Reference images are stored in a database, and live webcam captures are matched using Haar Cascade for detection and FaceNet for 128D encodings. Attendance is auto-updated in Excel upon a successful match. Tested in classrooms, the system showed highly satisfactory results.

III. METHODOLOGY

This work gets done in the following 5 stages:

1.Face Capturing: Student faces can be captured using various devices, including mobile phones, tablets, and CCTV cameras. When using a mobile device, the image is captured directly using the software and uploaded to the facial recognition server for attendance marking. In the case of CCTV cameras, live video feeds are processed in real time, analysing frames per second to identify students. In the event of a power outage, faculty members can manually capture snapshots of the class and upload them to the software, ensuring uninterrupted attendance tracking.

2.Face detection: The image or frame from a video is fed into the trained YOLOv10 model. The model processes the image, extracting relevant features indicative of face. YOLOv10 predicts bounding boxes around detected faces, providing their co-ordinates within the image. Each bounding box is assigned a confidence score, indicating the likelihood that the detected object is a face.

3.Face alignment: The system detects a face in an image using MTCNN. The detected face is analyzed to locate five key points:

- Left Eye
- Right Eye
- Nose
- Left Mouth Corner
- Right Mouth Corner

These points are used to determine the face orientation. Using the extracted landmarks, the face is aligned to a standard template. Affine transformation (or similarity transformation) is

applied to adjust rotation, scale, and translation so that the eyes and mouth appear in a fixed position. This ensures that all faces, regardless of how they were originally captured (tilted, rotated, or off-centre), are aligned similarly.

Benefit Of Combining YOLOV10 & MTCNN:

Combining YOLOv10 for detection and MTCNN for alignment brings the best of both worlds.

- **Higher Recognition Accuracy:** Better alignment => better embeddings => better recognition.
- **Reduced False Positives:** YOLOv10 gives precise face boxes; MTCNN ensures only valid faces are aligned and processed.
- **Real-Time Performance:** The combination is fast enough for real-time face recognition applications like attendance systems, security, surveillance, etc.
- **Handles Complex Scenarios:** Works well with crowded images, different poses, and lighting conditions.

Alternative (Why Not Use MTCNN Alone?)

MTCNN can detect and align faces on its own, but it's slower for detection, especially with large images or many faces. YOLOv10 is optimized for detection, so it's better to let YOLO do the heavy lifting for initial detection, then MTCNN focuses on fine alignment.

4.Face Recognition: The detected face is cropped and resized (e.g., 160×160 pixels). FaceNet passes the aligned face through a Convolutional Neural Network (CNN) to extract unique facial features. The output is a 128-dimensional embedding vector, which uniquely represents the face in a numerical format. Once the embeddings are generated, they are compared using a distance metric.

5. Classification: The classification process is carried out using SVM and KNN to identify the class to which the face belongs.

FLOWCHART:

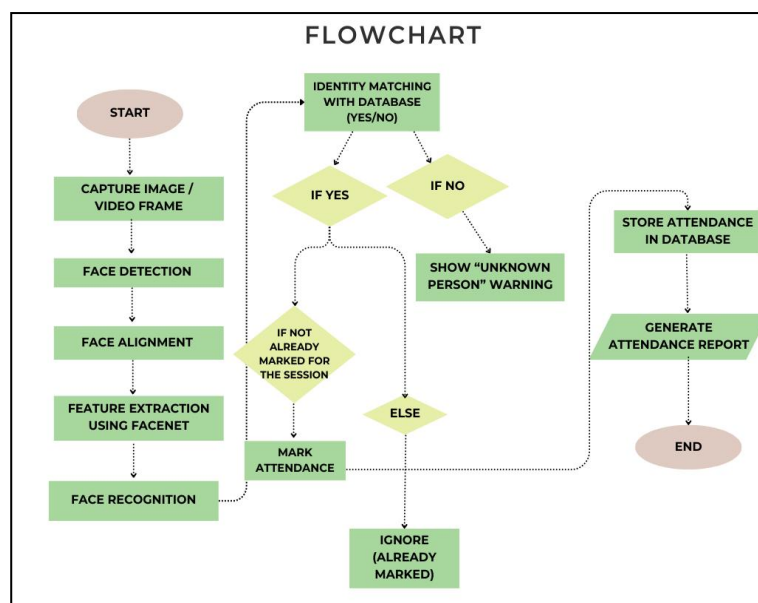
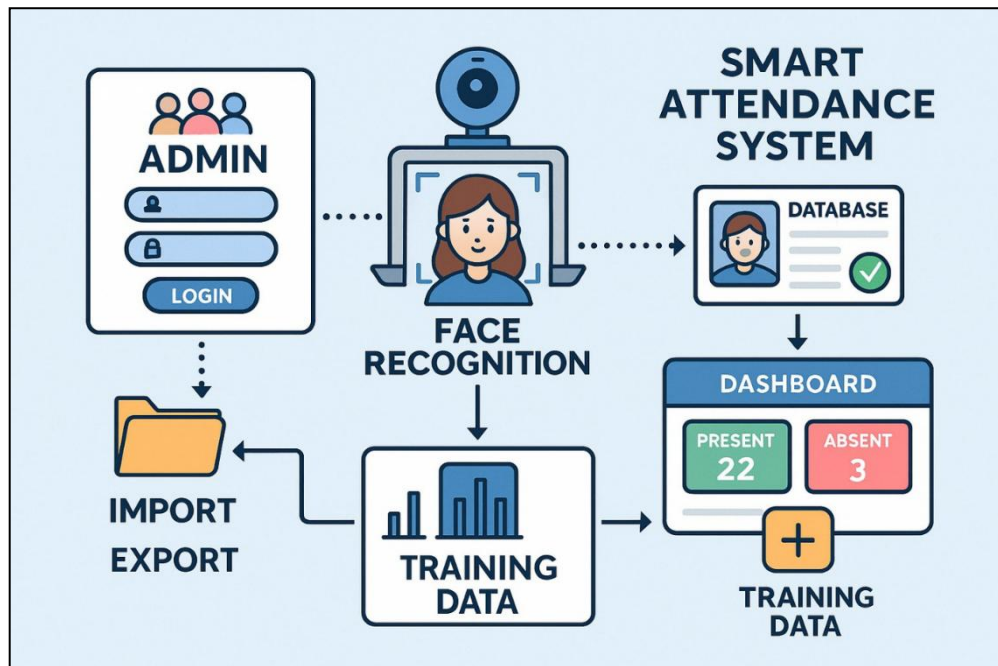


Fig. 1: Flowchart of the Smart Attendance System

SYSTEM ARCHITECHTURE:



1. Admin Login: The system begins with an admin login interface, where authorized personnel can access the attendance records and system settings. Administrators are required to enter their credentials, such as a username and password, to ensure security. Once logged in, they can manage users, update records, and configure the system.

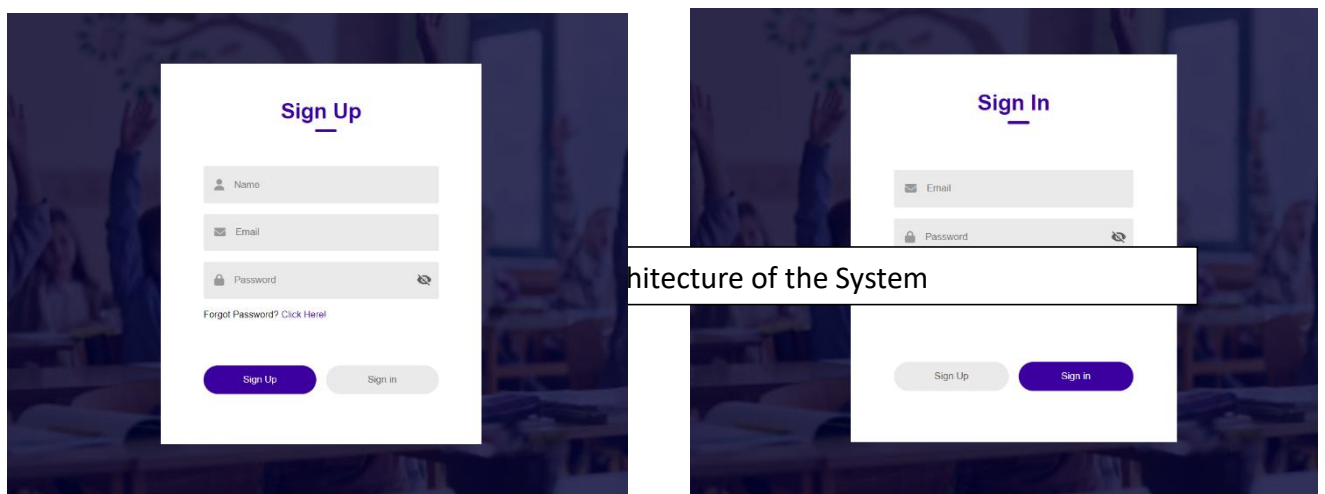


Fig. 3: Login Page

2. Face Recognition: At the core of the system is face recognition technology, which allows the system to identify and verify individuals based on their facial features. A camera or the CCTV captures the image of a person and compares it with pre-stored images in the database. If a match is found, the person's attendance is recorded automatically.

3. Database Management: The database stores the information of registered users, including their facial data and attendance history. When a person's face is scanned, the system checks against this database to verify their identity. If successfully recognized, the system marks them as present; otherwise, they remain absent.

	student_id [PK] character varying	faculty_id [PK] character varying	roll_number character varying	department character varying	year character (1)	subject_name character varying	status character varying	attendance_time time with time zone	student_name character varying
1	-	fac01	-	ECE	-	Mathematics	present	02:36:53.741381+05:30	Sudip Dogra
2	14200323019	-	19	ECE	2	Mathematics	present	02:36:53.741381+05:30	Debolina Saha
3	14200323024	-	24	ECE	2	Mathematics	absent	[default]	Shivani Dubey
4	14200323030	-	30	ECE	2	Mathematics	present	02:36:53.741381+05:30	Prarambhika Bhattachary
5	14200323040	-	40	ECE	2	Mathematics	present	02:36:53.741381+05:30	Koushik Ghoshal
6	14200323041	-	41	ECE	2	Mathematics	absent	[default]	Sreeja Malakar
7	14200323043	-	43	ECE	2	Mathematics	absent	[default]	Shubhrakamal Saha
8	14200323045	-	45	ECE	2	Mathematics	present	02:36:53.741381+05:30	Koushiki Ghosh
9	14200323053	-	53	ECE	2	Mathematics	absent	[default]	Raju Roy
10	14200323056	-	56	ECE	2	Mathematics	asbent	[default]	Farhan Iqbal

Fig. 4: Database

4. Dashboard and Attendance Tracking

The system provides a dashboard where attendance statistics are displayed in a clear and organized manner. The dashboard typically shows the number of present and absent individuals, helping administrators quickly assess attendance records.

5. Training Data for Face Recognition

The system improves accuracy by training on multiple facial images collected over time. More data enhances recognition, reducing false matches and ensuring reliable performance under varying conditions like lighting and expressions.

6. Import and Export of Data: The system also supports importing and exporting attendance records. This feature allows administrators to download attendance reports, share data with other systems, or upload new datasets for analysis. It facilitates smooth integration with external applications, such as payroll systems or academic management software, making it versatile for use in schools, offices, or other institutions.



Fig. 5: Import/Export Feature of the System

VI. RESULTS & DISCUSSION:

This section presents the outcome of the smart attendance system, analysing its performance and accuracy.

FACIAL RECOGNITION FROM VIDEO FRAMES

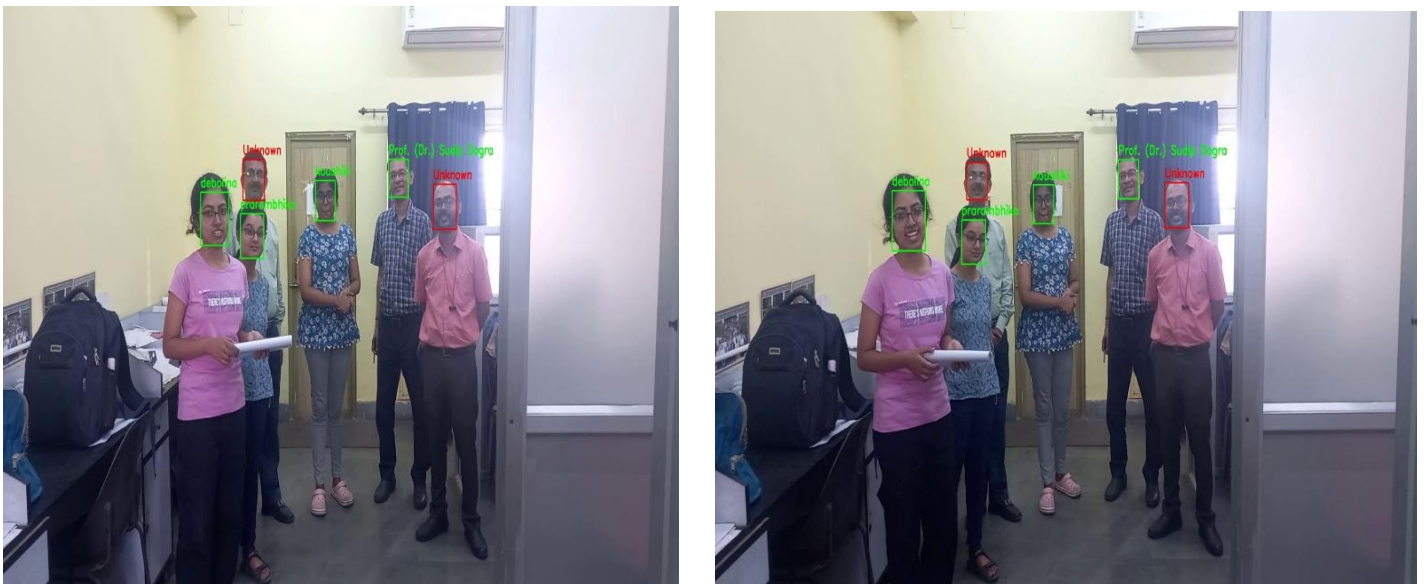


Fig. 6: Faces Detected and Recognized from video frames

The above image demonstrate face recognition from frames extracted from a video. This validates the architecture of our facial recognition system, as the video frames are processed through the face recognition server.

FACIAL RECOGNITION PERFORMED ON MASKED IMAGES

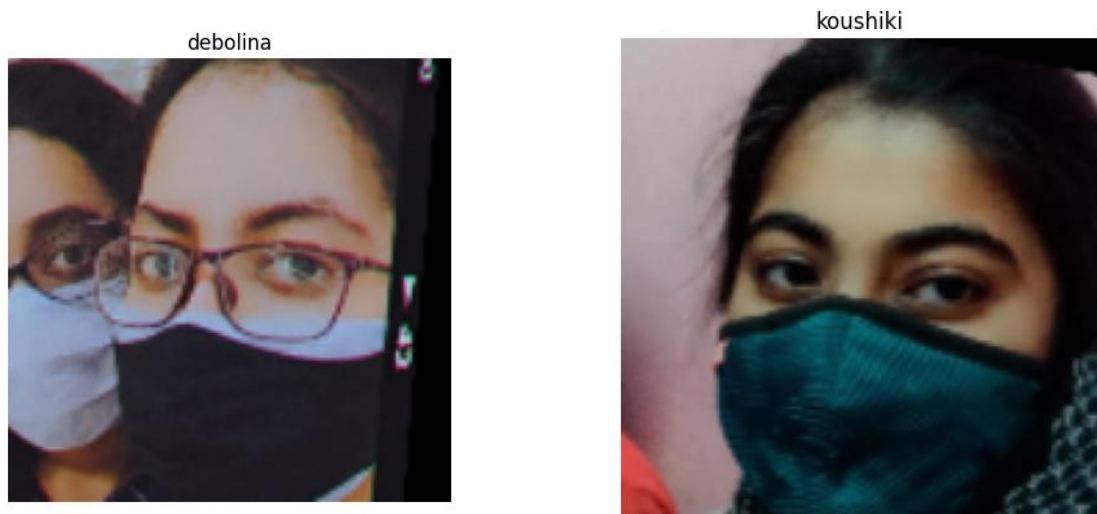


Fig. 7: Faces Recognized from Masked Images

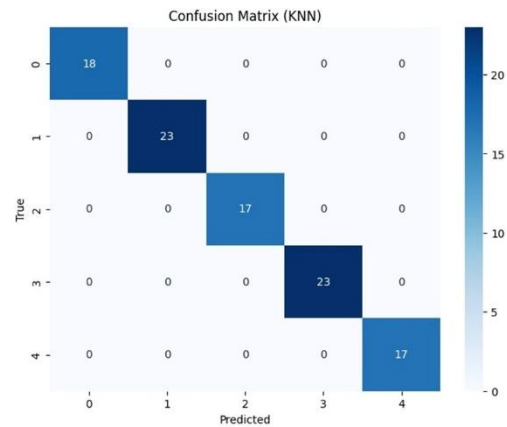
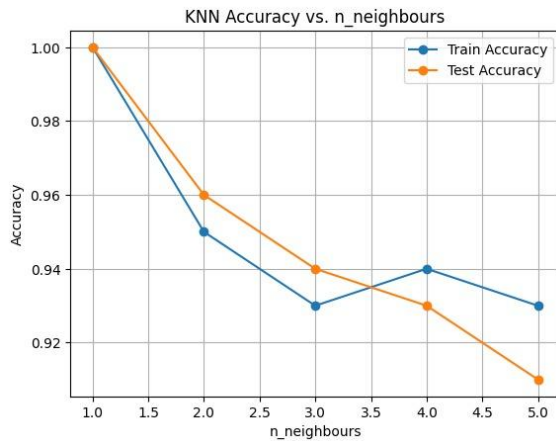
This facial recognition technique is designed to identify individuals even when their faces are partially covered by masks. The process involves detecting faces in video frames and using machine learning models to analyze visible facial features, such as the eyes and forehead, for identification. This feature makes it highly useful to recognize even partially covered faces.

SUCCESSFUL FACIAL RECOGNITION OF INDIVIDUALS AT A DISTANCE



The above two images demonstrate how the model accurately detects and identifies a person standing at a long distance, which is a common scenario in large classrooms where students are seated far from the camera.

MODEL ACCURACY:



The model's accuracy is inferred from the confusion matrix after selecting the optimal number of neighbors for KNN. The best value of n_neighbours is determined by plotting a graph for different n_neighbours values. This optimal value is then used to train the model, and the resulting confusion matrices provide insights into its accuracy. The model's accuracy in this evaluation was found to be 99%–100%, confirming that it successfully accomplishes the desired task.

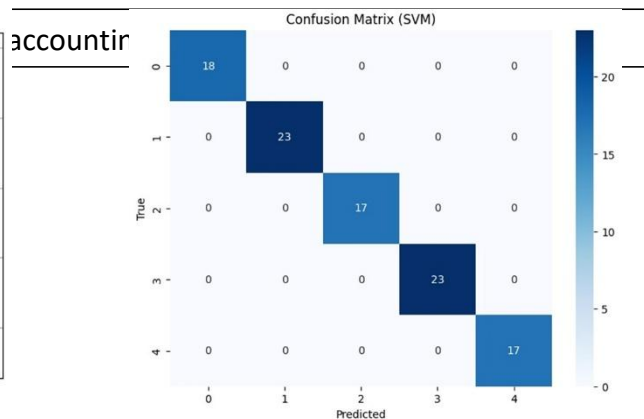
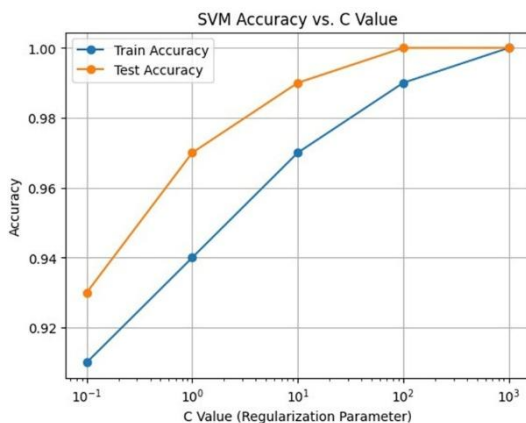


Fig. 10: Graphs accounting for model accuracy.

The model's accuracy is inferred from the confusion matrix after selecting the optimal regularization parameter (C) for SVM. The best C value is determined by plotting a graph for different C values. This optimal value is then used to train the SVM model, and the resulting confusion matrices provide insights into its accuracy. The model's accuracy in this evaluation was found to be 97%–98%, confirming that it successfully accomplishes the desired task.

CONCLUSION:

The Smart Attendance System using Face Recognition successfully automates the attendance process, reducing manual effort and minimizing errors associated with traditional methods. By leveraging machine learning and computer vision, the system accurately detects and identifies students, even in challenging conditions such as partial face occlusions, varying lighting, and long distances from the camera.

The performance evaluation, based on KNN and SVM classifiers, demonstrated high accuracy levels, with 99%–100% for KNN and 97%–98% for SVM, validating the system's reliability. The confusion matrix analysis further confirmed the effectiveness of the model in real-world classroom environments.

This system enhances efficiency, security, and scalability, making it a practical solution for educational institutions.

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REFERENCES:

1. **FaceNet: A Unified Embedding for Face Recognition and Clustering** by Florian Schroff, Dmitry Kalenichenko, and James Philbin of Google Inc.
Published in arXiv and IEEE Computer Society Conference on Computer Vision and Pattern Recognition (2015)
DOI: 10.48550/arXiv.1503.03832
2. **An Embedded Intelligent System for Attendance Monitoring** by TOUZENE Abderraouf, ABED Abdeljalil Wassim, Slimane LARABI
DOI: 10.48550/arXiv.2406.13694
3. **Design of an E-Attendance Checker through Facial Recognition using Histogram of Oriented Gradients with Support Vector Machine** by Allan Jason C. Arceo, Renee

Ylka N. Borejon, Mia Chantal R. Hortinela, Alejandro H. Ballado Jr., Arnold C. Paglinawan from School of EECE, Mapua University, Manila, Philippines

For the 8th International Conference on Smart City and Informatization (iSCI) [2020]

DOI: 10.1109/ISCI50694.2020.00008

4. Automating attendance management in human resources: A design science approach using computer vision and facial recognition by Bao-Thien Nguyen-Tat, Minh-Quoc Bui, Vuong M. Ngo for the International Journal of Information Management Data Insights 4 (2024) 100253

Publication : Elsevier | **DOI :** 10.1016/j.jjime.2024.100253

5. Joint Face Detection and Alignment using Multi-task Cascaded Convolutional Networks by Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, Senior Member, IEEE, and Yu Qiao, Senior Member, IEEE

Published in: IEEE Signal Processing Letters (Volume: 23, Issue: 10, October 2016)

Page(s): 1499 - 1503

DOI: 10.1109/LSP.2016.2603342

6. Dual-Proxy Modeling for Masked Face Recognition

Authors: Wang Shuhua, Mao Xiaochena

Published by Elsevier B.V. [2022]

Page(s): 145–151

7. Anti-spoofing-enabled Contactless Attendance Monitoring System in the COVID-19 Pandemic by Deepti Saraswata, Pronaya Bhattacharyaa, Tirth Shaha, Rushi Satania, Sudeep Tanwara published for the International Conference on Machine Learning and Data Engineering (2023)

Page(s): 1506-1515

8. A Real-Time Attendance System Using Deep Learning Face Recognition by Weidong Kuang, and Abhijit Baul, University of Texas Rio Grande Valley

Published in the year 2020